RSA and Period Finding:
Foundations, Resonance Formulation, Limits

Contents

Summary

This document treats the RSA procedure in two parts. Part I presents the number-theoretic foundations and security analysis. Part II examines the attack surface via period finding as a resonance problem and determines its carriers and limits: the resonance description is correct – period finding is a spectral problem – but its generators are the **primes**, not an adjustable parameter.

Two separate parameters

This document uses two quantities that must not be confused:

- $\xi = 4/30000$ – the FFGFT geometry parameter. Fixed, not adjustable, with no operational connection to factorisation.
- σ – a numerical grid parameter of the search algorithms (values such as $1/42$, $1/100$, $1/1000$). A sampling quantity, not a physical one.

Part I: The RSA Procedure

.1 Introduction to RSA Cryptography

The RSA algorithm (Rivest, Shamir, Adleman 1977) is one of the first practical public-key cryptosystems and is widely used for secure data transmission.

Mathematical Foundation

RSA is based on the computational difficulty of factoring large integers and the properties of modular arithmetic.

Key Generation

1. Choose two distinct prime numbers p and q.
2. Compute $n = p \cdot q$.
3. Compute Euler's totient: $\varphi(n) = (p-1)(q-1)$.
4. Choose e with $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$.
5. Compute d with $d \cdot e \equiv 1 \pmod{\varphi(n)}$.
   The public key is (n, e), the private key (n, d).

Encryption and Decryption

For a message M with $0 \leq M < n$:

$$C \equiv M^e \pmod{n} \qquad \text{(encryption)}, \tag{1}$$
$$M \equiv C^d \pmod{n} \qquad \text{(decryption)}. \tag{2}$$

.2 Mathematical Proofs

Correctness Proof

Theorem .2.1 (RSA Correctness)**.** *For any message M with $0 \leq M < n$ and $\gcd(M, n) = 1$: $(M^e)^d \equiv M \pmod{n}$.*

Proof. From $ed \equiv 1 \pmod{\varphi(n)}$ follows $ed = 1 + k\varphi(n)$. By Euler's theorem $M^{\varphi(n)} \equiv 1 \pmod{n}$, so

$$C^d \equiv M^{ed} \equiv M^{1+k\varphi(n)} \equiv M \cdot (M^{\varphi(n)})^k \equiv M \pmod{n}.$$

For $\gcd(M, n) \neq 1$ the Chinese Remainder Theorem gives the same. □

.3 Implementation

Modular Exponentiation

Efficient modular exponentiation by square-and-multiply:

Algorithm: Modular Exponentiation

Function ModExp($base$, $exponent$, $modulus$):

1. $result \leftarrow 1$; $base \leftarrow base$ mod $modulus$
2. **while** $exponent > 0$:
   (a) **if** $exponent$ mod $2 = 1$: $result \leftarrow (result \times base)$ mod $modulus$
   (b) $exponent \leftarrow \lfloor exponent/2 \rfloor$; $base \leftarrow (base \times base)$ mod $modulus$
3. **return** $result$

Prime Generation

- Probabilistic primality tests (Miller-Rabin).
- p and q of similar bit length.
- Avoid special forms that are easier to factor.

.4 Security Analysis

Common Attacks

Attack	Description
Factorisation	Factor n into p and q
Small e	Certain messages recoverable
Timing attacks	Deduce secret from computation time
Side-channel	Power, electromagnetic leaks

Recommendations

1. Key sizes ≥ 2048 bit (3072 or 4096 for long-term security).
2. Padding scheme OAEP.
3. Constant-time implementations.
4. Regular updates to cryptographic libraries.

.5 Performance Analysis

Operation	Complexity	Typical time (2048 bit)
Key generation	$O(k^3)$	1–10 seconds
Encryption	$O(k^2)$	$<$ 1 ms
Decryption	$O(k^3)$	10–100 ms

Optimisations

- Chinese Remainder Theorem for faster decryption.
- Windowing methods for exponentiation.
- Hardware acceleration.

.6 Extensions

Multi-Prime RSA

$n = p_1 p_2 \cdots p_k$: faster decryption via multi-prime CRT.

Blinding

Protection against timing attacks:

$$C' = C \cdot r^e \pmod{n}, \quad M' = (C')^d \pmod{n}, \quad M = M' \cdot r^{-1} \pmod{n}.$$

Part II: Period Finding as a Resonance Problem

.7 The arithmetic foundation

Factorisation of $n = pq$ reduces to period search. For random a with gcd$(a, n) = 1$ the period r satisfies $a^r \equiv 1 \pmod{n}$. If r is even and $a^{r/2} \not\equiv -1 \pmod{n}$ then gcd$(a^{r/2} \pm 1, n)$ yields the factors.

[K] The period r always divides $\lambda(n) = \mathrm{lcm}(p-1, q-1)$. It is an arithmetic quantity, not determined by a choosable parameter.

.8 The spectral formulation

Why period finding is a resonance problem

The sequence $f(x) = a^x \bmod n$ is periodic with period r. Its Fourier transform has sharp maxima at k/r. Period search is formally spectral analysis.

Primes as generators

[K] The resonance structure rests on primes – measured by the bicoherence test on 2469 Riemann zeros (Doc. 316, August 2026): coupling at prime powers ($b^2 = 0.76$–0.85), none at composites ($b^2 = 0.018$–0.025), contrast factor 30–50.

What the grid parameter σ achieves

[X] All σ values have composite denominators and form mixed points without their own resonance line. Their function is that of a sampling grid.

.9 Topology: winding number instead of frequency ratio

[K] For closed trajectories the winding number $W = \log_2(r)$ decides, not r. Pure fifth: $W = 0.5850$ (irrational) → open. Tempered: $W = 7/12$ (rational) → closed after 12 steps. Closure is an artefact of rationalisation.

.10 Structural scaling limit

The Weyl obstruction

[K] The Laplace spectrum of every compact manifold grows as $N(\lambda) \sim C_d \lambda^{d/2}$; the arithmetic counting density as $N(T) \sim T \ln T$ – not a power. A compact resonance space cannot carry a spectrum of density $T \ln T$.

Delimitation. This obstruction does not apply to Shor's algorithm. Shor seeks one period of one concrete number (finite task, $2n + 3$ qubits). What limits Shor at large n is the gate count ($\sim n^3$) and error-correction overhead – a *resource limit*.

The precision barrier

For sharp separation at $r \approx n$ a resolution $\Delta f \approx f_0/n^2$ is needed – at RSA sizes below any available number representation.

.11 Comparison of methods

Method	Principle	Closure	Limit
Trial division	Division	Discrete arithmetic	$O(\sqrt{n})$
Pollard ρ	Cycle detection	Birthday paradox	$O(n^{1/4})$
Resonance filter	Spectral search	Rationalised grid	$O(n)$
Shor's QFT	Quantum interference	Discrete phase	resource limit

Preparation (modular exponentiation) accounts for approximately **95 %** of the gate cost; the QFT for only 5 % (Doc. 176, August 2026).

[X] A general exponential quantum advantage is not established.

.12 What the Fourier transform achieves

[K] The Fourier transform is a *projection*: it translates a representation without generating new information. Measured on the Riemann zeros (Doc. 316): peaks at $\ln 2, \ln 3, \ln 5, \ln 7, \ln 11, \ln 13$ where information is present; no peaks at composite arguments.

.13 Balance

Statement	Status
RSA correctness (Euler, CRT)	**[K]** proved
Period finding is a spectral problem	**[K]**
Primes are the generators	**[K]** measured by bicoherence
Winding number decides closure	**[K]**
Preparation $\approx 95\,\%$ of gate cost	**[K]**
σ grids generate resonances	**[X]** sampling artefact
$\xi = 4/30000$ contributes to factorisation	**[X]** geometry
Classical period search $O((\log n)^3)$	**[X]** refuted
Weyl obstruction limits Shor	**[X]** resource limit
Fourier transform generates information	**[X]** projection

.14 Verification

Scripts in `2/python/Shor_Skripte/` (standard library, assertions, seed 20780458): `s1_periodensuche.py`, `s2_grenzen.py`, `s3_thermik_zustandsraum.py`.

.15 Relation to the corpus

Doc. 316 (August 2026): Weyl obstruction, bicoherence test, winding number, Fourier transform as projection. Revision declared under R75 in Doc. 190.